HYPERSTEER (Cross Attention)	Held-in	Held-out
Initialization		
Random	0.601	0.582
Pretrained	0.733	**0.623**
Decoder Blocks		
$N = 2$	0.707	0.549
$N = 4$	0.713	0.597
$N = 8$	0.721	0.610
$N = 20$	0.733	**0.623**
HYPERSTEER (No Context)		
Training Objective		
Reconstruction Loss	0.511	0.375
Language Modeling Loss	0.517	**0.340**

Table 2: Ablation study using `Gemma-2-2B` on architecture choices of HYPERSTEER after 1 epoch of training and evaluation on a small test set `Concept10`. We find that pre-trained initialization of the cross-attention architecture improves performance in both held-in and held-out scenarios, and in both cases performance improves with the number of hypernetwork decoder blocks. For the no-context hypernetworks which do not condition steering vectors on input prompts, reconstructing ground truth vectors is comparable to end-to-end training with a language modeling objective.

as we increase the training data used. We compare against the state-of-the-art supervised activation steering method `ReFT-r1`. Observe in Figure 3), that as training data increases HYPERSTEER becomes much more economical than supervised activation steering. See details in Appendix A.3.2.

Ablation Study We show results for various ablation studies on the cross-attention variant of HYPERSTEER in Table 2. We randomly initialize the `Gemma2-2B` hypernetwork and find that pretrained parameters provide a significant performance boost (+0.112 steering score). We also remove a number of hypernetwork decoder blocks in the range $N = \{2, 4, 8, 20\}$ and adjust learning rate accordingly: $\mathrm{lr}(n) = 8 \times 10^{-5} \cdot \sqrt{\frac{20}{n}}$. Increased depth results in incremental improvements to steering performance on held in and held out test sets. However, notably the number of decoder blocks has a greater impact on generalization to steering prompts unseen in training (+0.07) compared to steering prompts unseen in training (+0.03).

We also perform an ablation on the no context HYPERSTEER variant where we train the hypernetwork to reconstruct the steering vectors constructed by the original AxBench ReFT baselines. Given a steering vector $\mathcal{H}(s, x) = \mathcal{H}(s) = \Delta_s$ and a gold-label steering vector Δ_s^* the loss is

$$\mathcal{L}_{\mathsf{recon}}(s) = 1 - \mathsf{CosSim}(\Delta_s, \Delta_s^*) + ||\Delta_s - \Delta_s^*||_2^2.$$

The two loss terms are roughly comparable, so we use language modeling.

5 Qualitative Analyses

We generate 2500 steering vectors using base and steering prompts from our held-out test data.

Geometric visualization of steering vectors We analyze steering vectors generated by HYPERSTEER (Cross Attention) using t-SNE (van der Maaten and Hinton, 2008) and PCA (2 components) to find geometric structure among steering vectors (see Fig. 4 and 5 in App. A.7.1).

Pairwise similarity of steering vectors We compute pairwise cosine similarities of steering vectors on both in-context (reconstruction) and cross attention models to understand how conditioning on the input prompt affects semantics. The cross-attention variant (Figure 6a in App. A.7.1) yields high within-concept alignment but still shows off-diagonal similarities driven by shared prompt templates and linguistic structure. In contrast, the no-context variant (Figure 6b in A.7.1), conditioning on steering prompt only, produces much weaker off-diagonal alignment. We find that cross-attention's residual inter-concept similarity is weakened by this additional conditioning, but not at the cost of steering performance. Initial experiments to determine if geometric structure emerges among steering vectors sharing a concept yielded a negative. This is likely due to high semantic similarity of the prompts used in our evaluation pipeline.

6 Conclusion

Both held-in and held-out evaluations indicate that HYPERSTEER is a scalable and effective approach for steering language models. In particular, HYPERSTEER (Cross Attention), our best-performing variant, achieves significantly stronger performance on held-out prompts—improving further with dataset scale. It also outperforms all activation steering baselines on held-in evaluations. Without modifying model parameters, our method narrows the performance gap with fine-tuning and prompting. Finally, we demonstrate that HYPERSTEER becomes increasingly compute-efficient as data scale increases, achieving the same held-out loss with fewer training updates.

7 Limitations

Data A key limitation of our approach is the limited scope and quantity of the concept datasets. Using data with concepts of much greater complexity and difficulty from a model steering perspective would likely improve model performance and help make evaluation more robust. We also note that quality and robustness of concepts is bounded by the `GemmaScope` feature labels used to derive them, and collecting data from humans or other high quality sources is a feasible alternative. This is a key research priority we emphasize for future work.

Steering Sites All experiments in our work are limited to intervening on the *residual stream* activations of the base LM. There are other potentially more performant sites for intervention, including various points of the decoder block and during the attention computation. We also adopt the convention of prior work to intervene at all token positions; exploring more targeted interventions could reduce detrimental off-target steering effects and improve the overall steering score.

Compute Compared to supervised dictionary learning, the compute requirements of training a hypernetwork are large, as the number of trainable parameters significantly exceeds a `ReFT-r1`.

Model Scale Due to to compute constraints we only experimented with `Gemma-2-2B` architectures, which are worse instruction followers and in-context learners than the leading open source models with many more parameters. Training on models at a variety of scale would help cement HYPERSTEER 's strong steering performance against the improved in-context learning ability of larger LMs.

Open Source Models Our approach requires white-box access to a model's internals in order to use steering vectors, a limitation prompting does not encounter. Hence, we rely on the existence of sufficiently capable open source models as a basis for our research.

8 Ethical Considerations

We present this work with the intention that HYPERSTEER is a powerful tool for steering models away from producing harmful responses and better tailor outputs to downstream tasks. However, we acknowledge that model steering can also be used by bad actors as a tool to circumvent a target models's existing safety mechanisms or bias models towards misleading outputs or malicious persuasion. Hence, HYPERSTEER and hence steering vectors should be used responsibly and audited to prevent such issues from arising, and having a human-in-the-loop system could help mitigate some of these concerns.

9 Acknowledgments

AI Usage We use closed-source LLMs from OpenAI as a critical part of our work: synthetic concept data generation and evaluation pipelines utilize `gpt-4o-mini` to generate ground truth labels and judge responses according to criteria respectively.

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A Appendix

A.1 Future Directions

Large-scale concept data Prior works (Phang et al., 2023) explore pre-training the architecture